

An Adaptive Funnel-Based Stopping Strategy with No-Realized-Loss Exits

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Abstract

We propose a threshold-based stopping strategy for trading under drift and volatility. The method compares a log-price path against a drift-adjusted predictive funnel and exits a position only when three conditions are simultaneously satisfied: the price crosses the upper funnel boundary, local momentum weakens, and the log-return exceeds a transaction-cost-adjusted profit threshold. This gives a rigorous no-realized-loss property for completed algorithmic exits. The guarantee does not imply that the strategy is risk-free: mark-to-market wealth may decline while holding, capital may be locked for long periods, and forced exits may realize losses. The framework is formulated using discrete-time stochastic processes, stopping times, and elementary statistical estimation.

1 Introduction

A common trading intuition is to hold a stock while it is below or near the purchase price, and to sell only after a sufficiently favorable move once short-term upward momentum has weakened. The purpose of this paper is to formalize this idea as a rule-based stopping strategy rather than to claim global optimality.

The strategy combines three components:

1. a drift-adjusted deviation statistic measuring how far the log-price has moved relative to its expected path;
2. a local momentum statistic detecting whether recent upward movement has stalled;
3. an explicit profit threshold ensuring that completed algorithmic exits do not realize a loss after transaction costs.

The central point is that the strategy can impose a *no voluntary realized loss* constraint, but it cannot eliminate economic risk. While a position remains open, the mark-to-market value of the position can fall. Thus the strategy transforms realized-loss risk into holding-time risk, opportunity cost, and mark-to-market drawdown.

2 Price Model

Let $(S_t)_{t \geq 0}$ denote the asset price at discrete times. Define the log-price

$$X_t = \log S_t.$$

For the baseline model, assume

$$X_{t+1} = X_t + \mu + \sigma \varepsilon_{t+1}, \quad \varepsilon_{t+1} \sim \mathcal{N}(0, 1),$$

where $\mu \in \mathbb{R}$ is the drift and $\sigma > 0$ is the volatility. Let

$$\mathcal{F}_t = \sigma(X_0, \dots, X_t)$$

be the information available at time t .

3 Deviation Signal and Predictive Funnel

Fix a buying time τ_b . In the fixed-parameter baseline model, define the predictive center and cumulative variance by

$$C(t) = X_{\tau_b} + \mu(t - \tau_b), \quad V(t) = \sigma^2(t - \tau_b).$$

For a threshold parameter $k > 0$, define the upper and lower funnel boundaries by

$$U(t) = C(t) + k\sqrt{V(t)}, \quad L(t) = C(t) - k\sqrt{V(t)}.$$

Equivalently, define the normalized deviation statistic

$$Z_t = \frac{X_t - X_{\tau_b} - \mu(t - \tau_b)}{\sigma\sqrt{t - \tau_b}}, \quad t > \tau_b.$$

Then $X_t > U(t)$ is equivalent to $Z_t > k$.

Remark 1 (Fixed-time versus sequential interpretation). *For a fixed time t , under the Gaussian model, $Z_t \sim \mathcal{N}(0, 1)$. However, the trading rule monitors Z_t repeatedly over time. Therefore the relevant object is not only the fixed-time tail probability $P(Z_t > k)$, but the boundary-crossing probability*

$$P\left(\sup_{\tau_b < t \leq T} Z_t > k\right).$$

Thus k should be interpreted as a funnel-width or boundary-crossing tuning parameter, not simply as a fixed-time significance level.

4 Momentum Signal

For a window length $\delta \geq 1$, define the finite-difference momentum signal

$$M_t = X_t - X_{t-\delta}.$$

When $M_t > 0$, the recent log-price movement is upward over the last δ steps. When $M_t \leq 0$, recent upward momentum has stalled or reversed. Under the baseline Gaussian model,

$$M_t \sim \mathcal{N}(\mu\delta, \sigma^2\delta).$$

Hence M_t is a noisy local trend signal.

5 Transaction Costs and Profit Threshold

Suppose that buying incurs proportional cost $c_{\text{buy}} \geq 0$ and selling incurs proportional cost $c_{\text{sell}} \in [0, 1)$. Buying one unit at price S_{τ_b} effectively costs

$$(1 + c_{\text{buy}})S_{\tau_b},$$

and selling at price S_t yields

$$(1 - c_{\text{sell}})S_t.$$

To avoid a realized loss after transaction costs, it is necessary and sufficient that

$$(1 - c_{\text{sell}})S_t > (1 + c_{\text{buy}})S_{\tau_b}.$$

Taking logarithms gives the profit condition

$$X_t - X_{\tau_b} > h,$$

where

$$h = \log \left(\frac{1 + c_{\text{buy}}}{1 - c_{\text{sell}}} \right).$$

When transaction costs are ignored, $h = 0$.

6 Stopping Rule

Definition 1 (Adaptive funnel stopping rule, fixed-parameter version). *Given a buying time τ_b , define the selling time by*

$$\tau_s = \inf \{t > \tau_b : X_t > U(t), \quad M_t \leq 0, \quad X_t - X_{\tau_b} > h\}.$$

Equivalently,

$$\tau_s = \inf \{t > \tau_b : Z_t > k, \quad M_t \leq 0, \quad X_t - X_{\tau_b} > h\}.$$

The rule says: sell only after the price crosses the upper funnel boundary, local momentum has weakened, and the trade is profitable after costs.

7 Trading Cycle and Re-Entry Rule

The stopping rule above describes when an existing long position is exited. To turn the rule into a repeated trading strategy, we also specify how the strategy re-enters after a completed sale.

Suppose the strategy sells at time τ_s and then holds cash. During the cash period, take τ_s as the new reference time and define

$$Z_t^{\text{down}} = \frac{X_t - X_{\tau_s} - \mu(t - \tau_s)}{\sigma\sqrt{t - \tau_s}}, \quad t > \tau_s.$$

The next buying time is defined by the symmetric lower-funnel recovery rule

$$\tau_b^{\text{next}} = \inf \left\{ t > \tau_s : Z_t^{\text{down}} < -k, \quad M_t \geq 0 \right\}.$$

Thus the full algorithm alternates between two regimes:

1. while holding the asset, sell only when the upper-funnel, weakened-momentum, and profit-threshold conditions are all satisfied;
2. while in cash, buy after a lower-funnel deviation once local momentum has recovered.

The no-realized-loss theorem below applies to completed sell exits. The re-entry rule determines when capital is redeployed, but it does not by itself create a realized gain or loss.

8 No-Realized-Loss Property

Theorem 1 (No realized loss on completed algorithmic exits). *Suppose a trade is entered at time τ_b and exited at a finite time $\tau_s < \infty$ according to the stopping rule above. Then the net realized wealth multiplier satisfies*

$$\frac{(1 - c_{\text{sell}})S_{\tau_s}}{(1 + c_{\text{buy}})S_{\tau_b}} > 1.$$

Thus every completed algorithmic exit has positive realized return after transaction costs.

Proof. By the stopping rule,

$$X_{\tau_s} - X_{\tau_b} > h = \log \left(\frac{1 + c_{\text{buy}}}{1 - c_{\text{sell}}} \right).$$

Therefore

$$\frac{S_{\tau_s}}{S_{\tau_b}} = \exp(X_{\tau_s} - X_{\tau_b}) > \frac{1 + c_{\text{buy}}}{1 - c_{\text{sell}}}.$$

Multiplying both sides by $(1 - c_{\text{sell}})/(1 + c_{\text{buy}})$ gives

$$\frac{(1 - c_{\text{sell}})S_{\tau_s}}{(1 + c_{\text{buy}})S_{\tau_b}} > 1.$$

□

Remark 2. *This theorem applies only to completed algorithmic exits satisfying the stopping rule. If a finite horizon T is imposed and the position is forcibly liquidated at T , the theorem need not apply. Forced exits can realize losses.*

9 Wealth Processes

It is important to distinguish realized wealth from mark-to-market wealth. Let W_t^r denote realized wealth, updated only after completed exits. While holding a position purchased at τ_b , define mark-to-market wealth by

$$W_t^m = W_{\tau_b}^r \frac{S_t}{S_{\tau_b}}$$

in the frictionless case. With transaction costs interpreted as liquidation costs, one may instead use

$$W_t^m = W_{\tau_b}^r \frac{(1 - c_{\text{sell}})S_t}{(1 + c_{\text{buy}})S_{\tau_b}}.$$

The no-realized-loss theorem concerns W_t^r , not W_t^m . While holding, W_t^m may decrease.

A natural benchmark is buy-and-hold wealth,

$$W_t^{\text{bh}} = W_0 \frac{S_t}{S_0},$$

which should be compared against both realized and mark-to-market strategy wealth.

10 Momentum Decision Consistency

The finite-difference signal M_t is a noisy local trend statistic. A simple consistency result can be obtained by working with returns. Suppose that over a local window,

$$r_i = X_i - X_{i-1} = \beta + \epsilon_i,$$

where ϵ_i are independent with mean zero and finite variance. Define

$$\hat{\beta}_n = \frac{1}{n} \sum_{i=1}^n r_i.$$

Theorem 2 (Consistency of local mean-return decision). *Under the assumptions above,*

$$\hat{\beta}_n \xrightarrow{p} \beta.$$

Consequently, for any threshold $c \neq \beta$,

$$\mathbf{1}\{\hat{\beta}_n > c\} \xrightarrow{p} \mathbf{1}\{\beta > c\}.$$

Proof. The convergence $\hat{\beta}_n \xrightarrow{p} \beta$ follows from the law of large numbers. Since $c \neq \beta$, the indicator map is continuous at β , and the conclusion follows from the continuous mapping theorem. $\square$

This theorem justifies a smoothed momentum rule based on average recent returns. The simpler finite-difference signal $M_t = X_t - X_{t-\delta}$ corresponds to the cumulative return over the last δ steps and is noisier, but follows the same intuition.

11 Phase 2 Evaluation Metrics

The Phase 2 simulation evaluates the strategy by separating accounting quantities that can otherwise be confused. The baseline simulation tracks three wealth curves:

1. realized wealth W_t^r , which updates only after completed exits;
2. mark-to-market wealth W_t^m , which shows hidden drawdowns while holding;
3. buy-and-hold wealth W_t^{bh} , which measures opportunity cost.

From these curves, the main diagnostic quantities are:

$$R_T^r = \frac{W_T^r}{W_0} - 1, \quad R_T^m = \frac{W_T^m}{W_0} - 1, \quad R_T^{\text{bh}} = \frac{W_T^{\text{bh}}}{W_0} - 1.$$

The benchmark gap is measured by

$$G_T = \frac{W_T^m}{W_T^{\text{bh}}} - 1,$$

so $G_T > 0$ means the strategy's liquidating mark-to-market value exceeds buy-and-hold at the horizon, while $G_T < 0$ means the strategy underperforms buy-and-hold.

To measure hidden risk, define the mark-to-market drawdown process

$$D_t^m = \frac{W_t^m}{\max_{0 \leq s \leq t} W_s^m} - 1,$$

and its maximum drawdown over the horizon

$$D_{\min}^m = \min_{0 \leq t \leq T} D_t^m.$$

The same calculation can be applied to W_t^{bh} to compare the strategy's drawdown against the passive benchmark.

Finally, let $H_t \in \{0, 1\}$ denote whether the strategy is holding the asset. The exposure ratio

$$E_T = \frac{1}{T+1} \sum_{t=0}^T H_t$$

measures the fraction of time capital is exposed to the asset. Together, R_T^r , R_T^m , R_T^{bh} , G_T , $D_{\min}^m$, the number of completed trades, and E_T form the Phase 2 evaluation dashboard.

The simulation should use a Gaussian log-return model when validating the theory. If clipped shocks are used for visualization, they should be labelled as a visualization choice rather than as the exact model assumed in the paper.

12 Discussion and Limitations

The strategy enforces a no-realized-loss condition on completed voluntary exits. However, this does not imply that the strategy is risk-free. The main limitations are:

- **Mark-to-market drawdown:** the current value of the position may fall while holding;
- **Indefinite holding:** if the stopping rule never triggers, the position may remain open indefinitely;
- **Forced-exit risk:** finite-horizon liquidation can realize losses;
- **Opportunity cost:** the strategy may underperform buy-and-hold;
- **Model misspecification:** real prices have changing volatility, jumps, and fat tails;
- **Parameter estimation error:** the drift μ is difficult to estimate accurately.

Future phases should replace fixed μ and σ with estimated or adaptive quantities, such as Kalman-filtered drift and GARCH volatility, yielding an adaptive funnel.

13 Conclusion

We formulated a funnel-based stopping strategy for trading under drift and volatility. The key correction is the explicit profit threshold h , which gives a rigorous no-realized-loss theorem for completed algorithmic exits. The strategy remains risky in mark-to-market terms, and its practical value must be judged by simulation, buy-and-hold comparison, drawdown, and holding-time behavior.